

MADDIE POTTS

SOFTWARE ENGINEER | PYTHON, C++ & CUDA | SYSTEMS, HPC & SCIENTIFIC COMPUTING

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SUMMARY

Python-first software engineer and research scientist with strong C/C++ experience building shared libraries, distributed computing infrastructure, configuration-driven simulation software, automated analysis pipelines, hardware-integrated applications, and GPU-accelerated software. Technical lead for collaboration-wide projects with ownership spanning architecture, implementation, code review, testing, releases, reliability, and developer mentorship. Experienced in Linux, high-performance computing, REST APIs, reproducible scientific software, and translating complex technical requirements into maintainable systems.

TECHNICAL SKILLS

Programming: Python, C++, C, CUDA C++, Bash/Shell; working knowledge of JavaScript and SQL

Software Engineering: Software architecture, Git/GitHub, branching and pull requests, code review and merge approval, unit testing, debugging, release/version management, backwards compatibility, documentation, CMake, Make, Boost.Python

Systems & Infrastructure: Linux/Ubuntu, HPC, SLURM, distributed batch workflows, SSH, networking, remote operations, fault detection and recovery, Docker, Apptainer/Singularity

Data & Interfaces: REST APIs, HTTP, JSON, SQLite, database schema design, CSV, shared libraries, C++/Python interoperability

AI / ML: OpenAI API, LLM application development, PyTorch, convolutional neural networks

Scientific & GPU Computing: CERN ROOT, Geant4, CORSIKA, Monte Carlo simulation, CUDA kernels, shared-memory tiling, GPU performance benchmarking

PROFESSIONAL EXPERIENCE

PACIFIC NORTHWEST NATIONAL LABORATORY (PNNL) | Postdoctoral Research Assistant | May 2023 - Present

Department of Physics

- Lead and mentor 5 developers and contributors across multiple collaboration-wide SuperCDMS software projects, setting technical direction and defining software architecture and coding standards. Own feature planning, task assignment, code and pull-request reviews, merge approval, testing, validation, documentation, defect resolution, release readiness, and versioned releases.
- Reduced a several-hour manual analysis workflow to an essentially unattended process by designing and leading development of `cdmsproctools`, a multi-thousand-line Python 3 package that automatically generates and organizes 864 diagnostic plots per full run. The unit-tested software was adopted across the SuperCDMS collaboration and became a key analysis and diagnostic tool.
- Own the software lifecycle for `cdmsproctools`, managing Git-based collaborative development, code review and integration, unit testing, versioned releases, backwards compatibility, documentation, and contributions from other developers.
- Eliminated duplicated detector-configuration logic across the SuperCDMS software ecosystem by co-designing and leading development of Global Mappings, a modular system that centralized hundreds of detector and configuration parameters previously hard-coded independently across codebases. Built shared C++ libraries with Boost.Python bindings for consistent C++ and Python access; the system is distributed through collaboration containerized offline software releases.
- Independently designed and built a distributed computing orchestration system from the ground up that configured, submitted, and monitored 80,000 HPC jobs producing 500,000 simulations, implementing automated logging, failure detection, and multi-attempt recovery that detected and resubmitted failed workloads without manual intervention.
- Developed a reusable, configuration-driven C++/Geant4 detector-simulation framework supporting multiple research efforts, exposing geometry, materials, particle sources, positions, energies, physics settings, event counts, and output parameters through Geant4 macros and runtime commands. Validated detector response against Cs-137 and Ba-133 laboratory measurements.
- Designed and implemented a C++/SQLite detector-geometry data pipeline for a separate research project, defining the database schema and data format used by downstream reconstruction software and developing Geant4 code to traverse detector geometry, transform component positions into global coordinates, and generate the structured database.

- Developed the published low-energy calibration methodology and reusable Python/ROOT analysis software selected as the official calibration procedure for CUTE, leading calibration of 3 germanium detectors across 6 readout channels. Built reproducible event-selection, spectral-fitting, energy-calibration, characterization, visualization, and validation workflows that researchers can clone and independently execute.
- Work extensively in Linux development and scientific-computing environments; perform national-security research under a Department of Energy Q clearance.

PROFESSIONAL EXPERIENCE (CONTINUED)

GEORGIA INSTITUTE OF TECHNOLOGY | Postdoctoral Research Fellow | May 2022 - Oct. 2023

Trinity Neutrino Observatory and EUSO-SPB2

- Built the Trinity Demonstrator telescope from the ground up across software and hardware, driving procurement, assembly, integration, testing, field deployment, and operations. The deployed telescope actively collected scientific data in Utah.
- Designed and built the Trinity Demonstrator remote-control software architecture from the ground up, enabling unattended operation of the Utah-based telescope from Georgia Tech. Developed Python and Bash systems for data acquisition, hardware control, calibration, cameras, monitoring, automated data transfer, and authenticated REST API communication.
- Implemented automated fault detection for communication loss, timing failures, and power issues that alerted operators and placed the telescope in a safe state by closing the enclosure and disabling high voltage.
- Owned hardware procurement, Linux administration, and systems integration, deploying an industrial fanless Ubuntu PC for unattended operation at a remote mountain observatory. Configured networking, static IPs, SSH, packages, drivers, permissions, scheduled tasks, and automatic service recovery after power loss or reboot.
- Integrated a custom liquid-cooled detector system and coordinated research activities for 5 graduate and undergraduate researchers, organizing milestones, deadlines, technical discussions, and work required for field deployment and telescope operations.
- Led two remote field deployments and supported operation of computing, networking, data-acquisition, camera, calibration, and telescope-control hardware in a remote observatory environment.

TELESCOPE ARRAY COSMIC RAY OBSERVATORY | Researcher, University of Utah | Aug. 2015 - Sept. 2023

- Built the C++/CERN ROOT analysis framework that produced the final TAx4 cosmic-ray energy spectrum for my Ph.D. dissertation, developing custom data structures, file readers, event-selection algorithms, statistical fitting, automated visualization, and batch-processing workflows to analyze two years of detector data.
- Established the foundation for a machine-learning event-classification pipeline using Python and PyTorch, developing convolutional neural networks to distinguish photon- and hadron-induced cosmic-ray showers from detector data and Monte Carlo simulations; built training datasets, tuned hyperparameters, compared architectures, and implemented performance evaluation and confusion-matrix analysis.
- Served as primary administrator for a Linux research data server supporting 50+ users, managing accounts, permissions, storage, software, networking, backups, and system availability; diagnosed outages and restored service through remote and on-site troubleshooting while developing automation scripts supporting research workflows.
- Maintained and operated surface and fluorescence detector systems and performed Monte Carlo and ROOT-based analysis of detector response and cosmic-ray air-shower development.

SLAC AIR FLUORESCENCE EXPERIMENT (SFLASH) | Researcher, University of Utah

Aug. 2015 - May 2023

- Developed Bash scripts and analysis software supporting detector calibration, experimental data analysis, automated processing, and identification of high-quality beam runs.
- Supported 2016 and 2018 experiments at SLAC measuring air-fluorescence yield, including photomultiplier-tube calibration, beam-run data collection, experimental logging, and data-quality analysis.

SELECTED SOFTWARE PROJECTS

WARPTRACK | Independent Project | Python / PyTorch / C++ / CUDA | 2026 - Present

- Developing a CUDA-accelerated PyTorch event-classification framework for distinguishing muon-like particle tracks from hadronic cosmic-ray shower events using variable-length detector hit sets.
- Implemented a custom PyTorch C++/CUDA extension that computes batched pairwise squared-distance matrices between detector hits on the GPU, integrating CUDA kernels directly into the Python/PyTorch workflow.

- Built supporting software for synthetic event generation, variable-length padded batching and masking, model training, benchmarking, event inspection, and unit testing; configured the extension build process to run the test suite after compilation.

CUDA N-BODY SIMULATOR | Independent Project | C++ / CUDA | 2026 - Present

- Independently designed and developed a GPU-accelerated N-body gravitational simulator from scratch. Implemented custom CUDA kernels, host/device memory management, parallel force calculations, numerical integration, and interactive 3D visualization.
- Optimized memory access using shared-memory tiling and thread synchronization and benchmarked GPU performance with CUDA event timing across multiple thread/block and kernel launch configurations. Public source code is available through GitHub.

AI RESUME TAILORING APPLICATION | Independent Project | Python / OpenAI API | 2026 - Present

- Designed and developed an end-to-end LLM-powered job-application automation system that accepts a job-posting URL, retrieves and parses web content using Requests and BeautifulSoup, analyzes job requirements against resume content through the OpenAI API, generates tailored resumes and cover letters as Markdown, and automatically converts the results into finished PDF documents.

EDUCATION

Ph.D., Physics, University of Utah | 2022

M.S., Physics, University of Utah | 2021

B.S., Physics, University of Utah | 2017

SELECTED PUBLICATIONS

- M.F. Albakry et al. (SuperCDMS Collaboration). Calibration and Performance of Germanium High Voltage Detectors for SuperCDMS SNOLAB. 2026.
- A. Armatol et al. (TESSERACT Collaboration). Low Energy Phonon Bursts Created By Fast Neutron Damage. 2026.
- M. Bagheri et al. (Trinity Collaboration). The Camera and Readout for the Trinity Demonstrator and the EUSO-SPB2 Cherenkov Telescope. Nuclear Instruments and Methods in Physics Research, 2024.
- M. Potts. Ultra-High Energy Cosmic Ray Energy Spectrum Using Hybrid Analysis with TAx4. Ph.D. Dissertation, University of Utah, 2022.
- M. Potts and C. Jui (Telescope Array Collaboration). Monocular Energy Spectrum using the TAx4 Fluorescence Detector. Proceedings of Science, ICRC2021.
- R.U. Abbasi et al. (Telescope Array Collaboration). Surface detectors of the TAx4 experiment. Nuclear Instruments and Methods in Physics Research, 2021.

SELECTED PRESENTATIONS

- Simulation of BeEST Photopeak Calibration, APS Meeting, 2026, Poster.
- The SuperCDMS Experiment at SNOLAB, APS Meeting, 2026, Seminar.
- Searching for Dark Matter using SuperCDMS at SNOLAB, PoGo Symposium, 2024, Seminar.
- Progress on Trinity, an IACT Searching for UHE Neutrinos, APS April Meeting, 2023, Seminar.
- Trinity: UHE Earth-skimming Neutrino Detector, P5 Town Hall at Fermilab and Argonne, 2023, Seminar.
- Ultra-High Energy Cosmic Ray Energy Spectrum using Hybrid Analysis with TAx4, APS April Meeting, 2022, Seminar.

ADDITIONAL EXPERIENCE

UNIVERSITY OF UTAH | Teaching Assistant | Aug. 2017 - May 2019

- Created lesson plans and led in-class discussions in physics while helping students diagnose and overcome conceptual misunderstandings.

TOTALLY TUTORING | Contracted Tutor | May 2014 - June 2015

- Tutored physics, mathematics, and chemistry and created take-home assignments and short lesson plans.

ADDITIONAL TOOLS

Scientific/Engineering: MATLAB, Mathematica, LabVIEW, R, LaTeX, HTML/CSS

Working Environment: Linux package management, remote systems, storage and permissions, scientific hardware integration, laboratory and field deployments